In [1]:
```python
import sys, os
sys.path.insert(0, os.path.abspath('..'))

import torch
from src.data.data_loader import SatelliteDataLoader
from src.models.satellite_cnn import SatelliteCNN
from src.models.trainer import SatelliteCNNTrainer

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Using device: {device}')
```

```
Using device: cuda
```

In [2]:
```python
loader = SatelliteDataLoader(
    data_path="../src/data/UCMerced_LandUse/Images",
    batch_size=32
)

train_loader, val_loader = loader.get_loaders()
class_names = loader.get_class_names()

print(f'Classes: {len(class_names)}\nTraining batches: {len(train_loader)}\nValidat
```

```
Classes: 21
Training batches: 53
Validation batches: 14
```

In [3]:
```python
model = SatelliteCNN(num_classes=len(class_names))
print(f'Model created: {sum(p.numel() for p in model.parameters() if p.requires_gra
```

```
Model created: 24567893 trainable parameters
```

In [4]:
```python
trainer = SatelliteCNNTrainer(model, train_loader, val_loader, device=device)
results = trainer.train(epochs=2)

print(f'\nTest training completed with validation acc: {results["val_accs"][-1]:.2f
```

```
Training on device: cuda
Training samples: 1680
Validation samples: 420
----------------------

Epoch 1/2
Training: 100%|██████████| 53/53 [00:32<00:00,  1.64it/s, Loss=2.0536, Accuracy=48.7
5%]
Validating: 100%|██████████| 14/14 [00:17<00:00,  1.22s/it]
Train Loss: 2.0536, Train Accuracy: 48.75%
Val Loss: 0.6708, Val Accuracy: 83.57%
Learning Rate: 0.000100

New best model saved with Validation Accuracy: 83.57%!

Epoch 2/2
```

```
Training: 100%|██████████| 53/53 [00:29<00:00,  1.81it/s, Loss=0.4471, Accuracy=90.6
5%]
Validating: 100%|██████████| 14/14 [00:16<00:00,  1.19s/it]
Train Loss: 0.4471, Train Accuracy: 90.65%
Val Loss: 0.1523, Val Accuracy: 96.43%
Learning Rate: 0.000100

New best model saved with Validation Accuracy: 96.43%!

Training completed in 96s
Best Validation Accuracy: 96.43%

Test training completed with validation acc: 96.43%
```